Data Science Project Report

House Price Prediction Using Machine Learning

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Abstract

House price prediction is an important application of machine learning in the real estate sector. This project aims to predict residential property sale prices using the Kaggle House Prices dataset. Data preprocessing, exploratory data analysis, feature selection, and Linear Regression modeling were performed. The developed model achieved an R^2 score of 79.16%, demonstrating its ability to capture the relationship between property characteristics and house prices.

1. Introduction

Real estate valuation is a challenging task due to the influence of multiple factors such as property size, quality, location, and amenities. Machine learning techniques can automate the estimation process and provide data-driven insights. The objective of this project is to build a predictive model capable of estimating house sale prices using historical housing data.

2. Dataset Description

Source: Kaggle House Prices Dataset

Dataset Characteristics:

- Total Training Records: 1460
- Total Features: 81
- Target Variable: SalePrice

Important Features:

- OverallQual
- GrLivArea
- GarageCars
- GarageArea
- TotalBsmtSF

The dataset contains both numerical and categorical attributes describing residential properties.

3. Data Preprocessing

The preprocessing stage involved:

- Identification of missing values.
- Removal of columns with excessive missing data:
    - PoolQC
    - Fence
    - Alley
    - MiscFeature
- Median imputation for numerical variables.
- Mode imputation for categorical variables.
- Feature selection based on correlation analysis.

This step improved data quality and reduced noise.

4. Exploratory Data Analysis

Exploratory analysis was performed to understand the distribution of house prices and the relationship between features.

Key Findings:

- SalePrice follows a right-skewed distribution.
- OverallQual has the highest correlation with SalePrice.
- GrLivArea shows a strong positive relationship with house prices.
- GarageArea and TotalBsmtSF contribute significantly to property valuation.

Insert:

Actual vs Predicted Scatter Plot

5. Model Development

Model Used:

- Linear Regression

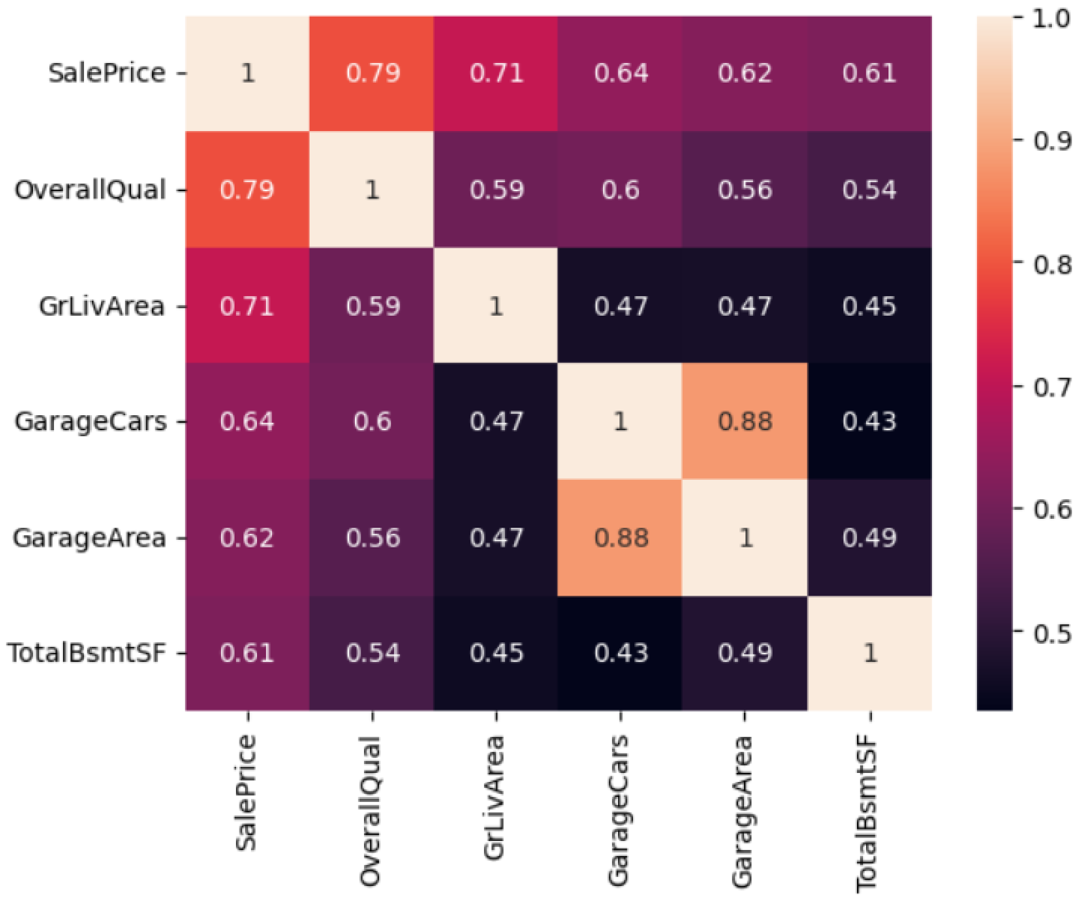
Workflow:

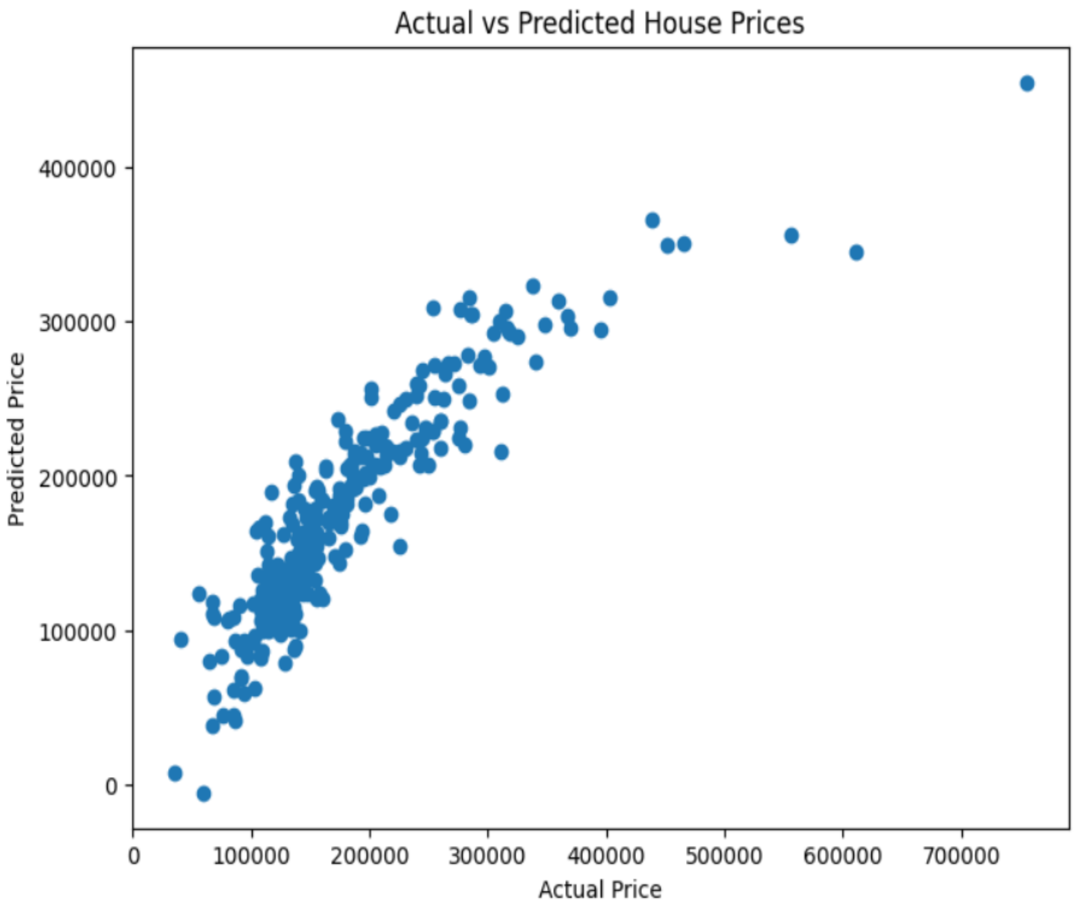
1. Feature Selection
2. Train-Test Split (80%-20%)
3. Model Training

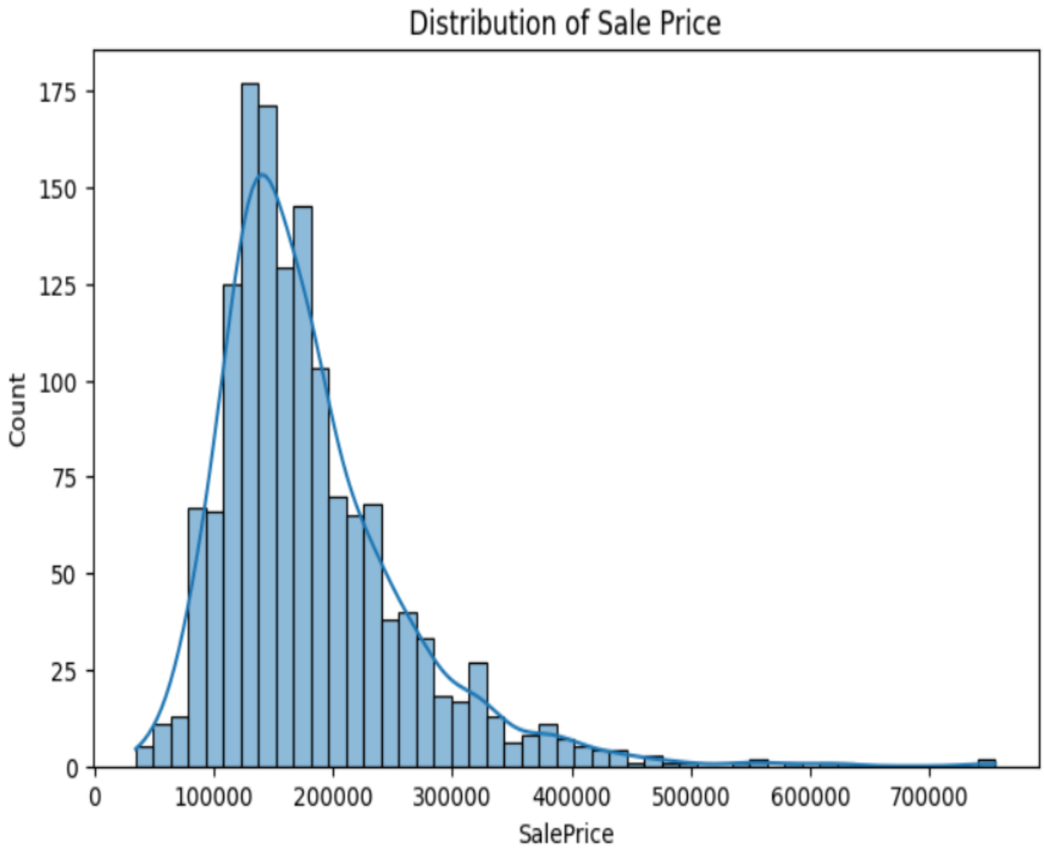
4. Prediction
5. Performance Evaluation

Selected Features:

- OverallQual
- GrLivArea
- GarageCars
- GarageArea
- TotalBsmtSF




Actual vs Predicted House Prices
Predicted Price
400000
300000
200000
100000
0
0
100000
200000
300000
400000
500000
600000
700000
Actual Price


Distribution of Sale Price
Count
175
150
125
100
75
50
25
0
0
100000
200000
300000
400000
500000
600000
700000
SalePrice

6. Results

Model Performance:

MAE = 25,284.81

RMSE = 39,979.43

R^2 Score = 0.7916

Interpretation:

The model explains approximately 79.16% of the variation in house prices. The results indicate that the selected features are effective predictors of residential property values.

Insert:
Actual vs Predicted Scatter Plot

7. Conclusion

This project successfully developed a machine learning model for house price prediction. Linear Regression provided satisfactory performance with an R^2 score of 79.16%.

The analysis revealed that overall quality, living area, garage capacity, garage area, and basement area are major factors influencing house prices.

8. Future Scope

- Random Forest Regression
- XGBoost Regression
- Hyperparameter Tuning
- Advanced Feature Engineering
- Deployment as a Web Application